

Geometry and Measures

GCSE Maths · Worked Solutions

20 questions · 62 marks total

jtmaths.org · free maths resources

Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Two angles lie on a straight line. One of them is 130° .

Work out the size of the other angle.

1 Angles on a straight line add up to 180° .

2 $180 - 130 = 50$

Answer: 50°

2 [2 marks]

(a) Work out the area of a triangle with base 10 cm and perpendicular height 6 cm.

(b) Work out the area of a parallelogram with base 7 cm and perpendicular height 4 cm.

1 (a) Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

2 $= \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$

3 (b) Area of a parallelogram = base $\times$ height (no halving).

4 $= 7 \times 4 = 28 \text{ cm}^2$

Answer: (a) 30 cm^2 (b) 28 cm^2

3 [2 marks]

Two angles of a triangle are 55° and 68° .

Work out the size of the third angle.

1 The angles in a triangle add up to 180° .

2 $55 + 68 = 123$

3 $180 - 123 = 57$

Answer: 57°

4 [2 marks]

A circle has radius 5 cm.

Work out the circumference of the circle, correct to 1 decimal place.

1 Circumference = $2\pi r$ (use the radius, not the diameter).

2 $= 2 \times \pi \times 5 = 10\pi$

3 $10 \times 3.14159... = 31.4159...$

Answer: 31.4 cm (1 d.p.)

5

[3 marks]

A circle has diameter 12 cm.

Work out the area of the circle, correct to 1 decimal place.

1 The area formula uses the radius, so halve the diameter first: $r = 12 \div 2 = 6$ cm

2 $\text{Area} = \pi r^2 = \pi \times 6^2$

3 $= 36\pi$

4 $36 \times 3.14159... = 113.097...$

Answer: 113.1 cm² (1 d.p.)

6

[3 marks]

Work out the size of one interior angle of a regular octagon.

1 The exterior angles of any polygon add up to 360°.

2 A regular octagon has 8 equal exterior angles: $360 \div 8 = 45^\circ$

3 An interior angle and its exterior angle lie on a straight line.

4 Interior angle = $180 - 45 = 135^\circ$

Answer: 135°

7

[3 marks]

A right-angled triangle has the two shorter sides 5 cm and 12 cm.

(a) Work out the length of the hypotenuse.

(b) Work out the perimeter of the triangle.

1 (a) Pythagoras' theorem: $a^2 + b^2 = c^2$

2 $5^2 + 12^2 = 25 + 144 = 169$

3 $c = \sqrt{169} = 13$ cm

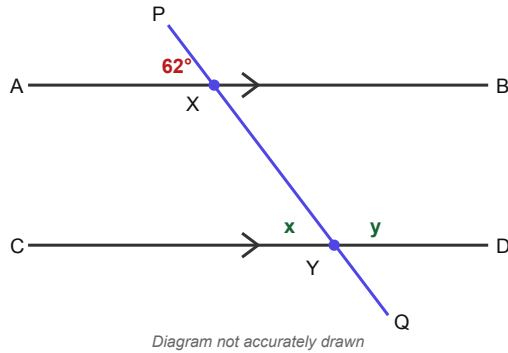
4 (b) Perimeter = $5 + 12 + 13 = 30$ cm

Answer: (a) 13 cm (b) 30 cm

8

[3 marks]

In the diagram, AB is parallel to CD. The line PQ crosses AB at X and CD at Y.



(a) Work out the size of the angle marked x . Give a reason for your answer.

(b) Work out the size of the angle marked y . Give a reason for your answer.

1 (a) The 62° angle and x are in matching positions at the two crossings — both sit above their line and to the left of PQ.

2 Corresponding angles on parallel lines are equal, so $x = 62^\circ$

3 (b) y and x together lie on the straight line CD, so they add to 180° .

4 $y = 180 - 62 = 118^\circ$

Answer: (a) $x = 62^\circ$ (corresponding angles are equal) (b) $y = 118^\circ$ (angles on a straight line add to 180°)

9

[4 marks]

A prism has a cross-section that is a right-angled triangle with shorter sides 6 cm and 8 cm. The prism is 15 cm long.

(a) Work out the area of the cross-section.

(b) Work out the volume of the prism.

(c) Work out the total surface area of the prism.

1 (a) Area = $\frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$

2 (b) Volume of a prism = area of cross-section $\times$ length = $24 \times 15 = 360 \text{ cm}^3$

3 (c) First find the third side of the triangle: $\sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$

4 The two triangular ends give $2 \times 24 = 48 \text{ cm}^2$

5 The three rectangles have a combined width equal to the triangle's perimeter: $6 + 8 + 10 = 24 \text{ cm}$

6 So the rectangles give $24 \times 15 = 360 \text{ cm}^2$, and the total is $48 + 360 = 408 \text{ cm}^2$

Answer: (a) 24 cm^2 (b) 360 cm^3 (c) 408 cm^2

10 [4 marks]

A shape is made from a rectangle 12 cm by 8 cm with a semicircle of diameter 8 cm cut out of one of the shorter ends.

Work out the area of the shape that is left, correct to 1 decimal place.

- 1 Work out the two areas separately and subtract.
- 2 Rectangle: $12 \times 8 = 96 \text{ cm}^2$
- 3 The semicircle has diameter 8 cm, so its radius is 4 cm.
- 4 Area of a full circle = $\pi \times 4^2 = 16\pi = 50.265... \text{ cm}^2$
- 5 A semicircle is half of that: $25.132... \text{ cm}^2$
- 6 Remaining area = $96 - 25.132... = 70.867... \text{ cm}^2$

Answer: 70.9 cm² (1 d.p.)

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1 [2 marks]

Work out the sum of the interior angles of a 12-sided polygon.

- 1 The sum of the interior angles of an n-sided polygon is $(n - 2) \times 180^\circ$.
- 2 Here $n = 12$, so $(12 - 2) \times 180 = 10 \times 180$
- 3 $= 1800^\circ$

Answer: 1800°

2 [2 marks]

A cylinder has radius 4 cm and height 10 cm.

Work out its volume, correct to 3 significant figures.

- 1 Volume of a cylinder = $\pi r^2 h$
- 2 $= \pi \times 4^2 \times 10 = 160\pi$
- 3 $160 \times 3.14159... = 502.65...$

Answer: 503 cm³ (3 s.f.)

3 [3 marks]

A right-angled triangle has hypotenuse 12 cm.

One of the other angles is 35°.

Work out the length of the side opposite the 35° angle, to 3 significant figures.

- 1 You know the hypotenuse and want the opposite side, so use sine (SOH).
- 2 $\sin 35^\circ = \frac{\text{opposite}}{12}$
- 3 Rearrange: opposite = $12 \times \sin 35^\circ$
- 4 $\sin 35^\circ = 0.573576...$, so opposite = $12 \times 0.573576 = 6.8829...$

Answer: 6.88 cm (3 s.f.)

4

[3 marks]

In a right-angled triangle the side opposite angle θ is 7 cm and the side adjacent to θ is 9 cm.
Work out the size of angle θ , correct to 1 decimal place.

1 Opposite and adjacent means tangent (TOA).

$$2 \quad \tan \theta = \frac{7}{9} = 0.7777\dots$$

3 To find the angle, use the inverse: $\theta = \tan^{-1}(0.7777\dots)$

$$4 \quad \theta = 37.874\dots^\circ$$

Answer: $\theta = 37.9^\circ$ (1 d.p.)

5

[4 marks]

(a) A, B and C are points on a circle with centre O. The angle AOC at the centre is 130° , and B lies on the major arc. Work out the size of angle ABC, giving a reason.

(b) PQRS is a cyclic quadrilateral. Angle PSR = 115° . Work out the size of angle PQR, giving a reason.

1 (a) The angle at the centre is twice the angle at the circumference when both stand on the same arc.

$$2 \quad \text{Angle ABC} = 130 \div 2 = 65^\circ$$

3 (b) Opposite angles of a cyclic quadrilateral add up to 180° .

$$4 \quad \text{Angle PQR} = 180 - 115 = 65^\circ$$

Answer: (a) 65° — the angle at the centre is twice the angle at the circumference (b) 65° — opposite angles of a cyclic quadrilateral add to 180°

6

[4 marks]

In triangle ABC, angle A = 42° , angle B = 68° and side a (opposite A) = 9 cm.

Work out the length of side b, correct to 3 significant figures.

1 You have two angles and the side opposite one of them, so use the sine rule.

$$2 \quad \frac{a}{\sin A} = \frac{b}{\sin B}$$

$$3 \quad \text{Substitute: } \frac{9}{\sin 42^\circ} = \frac{b}{\sin 68^\circ}$$

$$4 \quad \text{Rearrange: } b = \frac{9 \times \sin 68^\circ}{\sin 42^\circ}$$

$$5 \quad \sin 68^\circ = 0.927184 \text{ and } \sin 42^\circ = 0.669131$$

$$6 \quad b = \frac{9 \times 0.927184}{0.669131} = \frac{8.34465}{0.669131} = 12.470\dots$$

Answer: $b = 12.5$ cm (3 s.f.)

7

[4 marks]

A triangle has two sides of length 7 cm and 9 cm with an angle of 58° between them.

(a) Work out the length of the third side, to 3 significant figures.

(b) Work out the area of the triangle, to 3 significant figures.

1 (a) Two sides and the angle between them means the cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$$2 \quad c^2 = 7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 58^\circ$$

$$3 \quad = 49 + 81 - 126 \times 0.529919 = 130 - 66.769... = 63.230...$$

$$4 \quad c = \sqrt{63.230...} = 7.951... \text{ cm}$$

$$5 \quad \text{(b) Area} = \frac{1}{2}ab \sin C = \frac{1}{2} \times 7 \times 9 \times \sin 58^\circ$$

$$6 \quad = 31.5 \times 0.848048 = 26.713... \text{ cm}^2$$

Answer: (a) 7.95 cm (b) 26.7 cm² (both 3 s.f.)

8

[4 marks]

A sector of a circle has radius 9 cm and angle 140° .

(a) Work out the length of the arc, to 3 significant figures.

(b) Work out the area of the sector, to 3 significant figures.

1 A sector is a fraction $\frac{140}{360}$ of the whole circle.

$$2 \quad \text{(a) Full circumference} = 2 \times \pi \times 9 = 18\pi = 56.5487... \text{ cm}$$

$$3 \quad \text{Arc} = \frac{140}{360} \times 56.5487... = 21.991... \text{ cm}$$

$$4 \quad \text{(b) Full area} = \pi \times 9^2 = 81\pi = 254.469... \text{ cm}^2$$

$$5 \quad \text{Sector area} = \frac{140}{360} \times 254.469... = 98.960... \text{ cm}^2$$

Answer: (a) 22.0 cm (b) 99.0 cm² (both 3 s.f.)

9

[4 marks]

In triangle OAB, the vector from O to A is **a** and the vector from O to B is **b**.

M is the midpoint of AB. N is the point on OB such that $ON : NB = 1 : 2$

(a) Write the vector from A to B in terms of **a** and **b**.

(b) Write the vector from O to M in terms of **a** and **b**.

(c) Write the vector from M to N in terms of **a** and **b**.

1 (a) To go from A to B, travel back to O then out to B: $-\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$

2 (b) M is halfway along AB, so from O go to A then half of AB.

$$3 \quad OM = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$$

4 (c) $ON : NB = 1 : 2$ means N is $\frac{1}{3}$ of the way along OB, so $ON = \frac{1}{3}\mathbf{b}$

$$5 \quad MN = MO + ON = -OM + ON = -\frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{b} + \frac{1}{3}\mathbf{b}$$

$$6 \quad \text{Combine the } \mathbf{b} \text{ terms: } -\frac{1}{2} + \frac{1}{3} = -\frac{1}{6}$$

Answer: (a) $\mathbf{b} - \mathbf{a}$ (b) $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ (c) $-\frac{1}{2}\mathbf{a} - \frac{1}{6}\mathbf{b}$

10

[5 marks]

A cuboid ABCDEFGH measures 8 cm by 6 cm by 5 cm, where ABCD is the 8 cm by 6 cm base and AE = 5 cm is vertical.

(a) Work out the length of the diagonal AG that runs from one corner of the base to the opposite corner of the top. Give your answer to 3 significant figures.

(b) Work out the angle that AG makes with the base ABCD, to 1 decimal place.

1 (a) Work in two stages. First find the diagonal AC across the base using Pythagoras.

$$2 \quad AC^2 = 8^2 + 6^2 = 64 + 36 = 100, \text{ so } AC = 10 \text{ cm}$$

3 Now AG is the hypotenuse of the right-angled triangle ACG, where CG = 5 cm is vertical.

$$4 \quad AG^2 = AC^2 + CG^2 = 100 + 25 = 125, \text{ so } AG = \sqrt{125} = 11.180... \text{ cm}$$

5 (b) The required angle is GAC, inside that same right-angled triangle.

$$6 \quad \text{The side opposite is } CG = 5 \text{ and the side adjacent is } AC = 10, \text{ so } \tan(GAC) = \frac{5}{10} = 0.5$$

$$7 \quad GAC = \tan^{-1}(0.5) = 26.565...^\circ$$

Answer: (a) 11.2 cm (3 s.f.) (b) 26.6° (1 d.p.)